

Data Generation Execution Guide (Parquet + Delta + SCD2 + AI/ML/LLM)

This is a terminal-first runbook for the **data generation track**.

- Repo root: `/Users/natarajankanakasabapathy/FECTECH/TestDataGenerator`
 - Baseline config used here: `config/Acct_bkng.xlsx`
 - Relationship-inference demo config: `config/Creditcard_no_rel.xlsx`
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0) Memory Flow

L -> G -> D -> S

- Lint config
- Generate snapshots (v1, v2)
- Delta between snapshots
- SCD2 history from snapshots

For relationship inference + LLM:

I -> R -> F

- Infer relationships
 - Review inferred YAML
 - Feedback record (optional but recommended)
-

1) One-Time Setup

```
cd /Users/natarajankanakasabapathy/FECTECH/TestDataGenerator
poetry install
```

Optional extras (only if you need these features):

```
pip install matplotlib
poetry install --extras mimesis
```

2) Baseline Lint + Snapshot Generation

Step 1: Lint config

```
cd /Users/natarajankanakasabapathy/FECTECH/TestDataGenerator
poetry run python main.py lint --config config/Acct_bkng.xlsx
```

Step 2: Generate snapshot v1

```
cd /Users/natarajankanakasabapathy/FECTECH/TestDataGenerator
poetry run python main.py generate --config config/Acct_bkng.xlsx --output output/snap_v1 --
```

Step 3: Generate snapshot v2 (different seed so data changes)

```
cd /Users/natarajankanakasabapathy/FECTECH/TestDataGenerator
poetry run python main.py generate --config config/Acct_bkng.xlsx --output output/snap_v2 --
```

Step 4: Inspect parquet output (optional)

```
cd /Users/natarajankanakasabapathy/FECTECH/TestDataGenerator
poetry run python readParquet.py output/snap_v1
poetry run python readParquet.py output/snap_v2
```

3) Parquet Delta Execution

Step 1: Run delta between snapshots

```
cd /Users/natarajankanakasabapathy/FECTECH/TestDataGenerator
poetry run python main.py delta --config config/Acct_bkng.xlsx --previous output/snap_v1 --c
```

Step 2: (Optional) Delta for selected tables only

```
cd /Users/natarajankanakasabapathy/FECTECH/TestDataGenerator
poetry run python main.py delta --config config/Acct_bkng.xlsx --previous output/snap_v1 --c
```

Step 3: (Optional) Override partition column

```
cd /Users/natarajankanakasabapathy/FECTECH/TestDataGenerator
poetry run python main.py delta --config config/Acct_bkng.xlsx --previous output/snap_v1 --c
```

4) SCD2 Execution

Step 1: Build SCD2 history

```
cd /Users/natarajankanakasabapathy/FECTECH/TestDataGenerator
poetry run python main.py scd2 --config config/Acct_bkng.xlsx --previous output/snap_v1 --c
```

Step 2: Inspect SCD2 output (optional)

```
cd /Users/natarajankanakasabapathy/FECTECH/TestDataGenerator
poetry run python readParquet.py output/scd2_run
```

Expected SCD2 metadata columns include fields like: - effective_from_ts - effective_to_ts - is_current - version_num

5) AI/ML Relationship Inference (No API key required)

Step 1: Generate with inferred relationships (ML default)

```
cd /Users/natarajankanakasabapathy/FECTECH/TestDataGenerator
poetry run python main.py generate --config config/Creditcard_no_rel.xlsx --output output/in
```

Step 2: Standalone infer (reviewable YAML + ER diagram)

```
cd /Users/natarajankanakasabapathy/FECTECH/TestDataGenerator
poetry run python main.py infer-relationships --config config/Creditcard_no_rel.xlsx --confi
```

Step 3: Review + edit inferred YAML

- Open config/inferred_creditcard.yaml
- Keep/remove/add relationships as SME review output
- Save reviewed file as: config/inferred_creditcard.reviewed.yaml

Step 4: Record feedback for adaptive learning

```
cd /Users/natarajankanakasabapathy/FECTECH/TestDataGenerator
poetry run python main.py record-feedback --inferred config/inferred_creditcard.yaml --revie
```

6) LLM Relationship Inference + LLM Data Features

6.1 Export LM Studio environment

```
export SDP_LLM_PROVIDER="lm-studio"
export SDP_LLM_BASE_URL="http://localhost:1234/v1"
export SDP_LLM_MODEL="google/gemma-4-e4b"
```

6.2 Generate with LLM relationship inference

```
cd /Users/natarajankanakasabapathy/FECTECH/TestDataGenerator
poetry run python main.py generate --config config/Creditcard_no_rel.xlsx --output output/in
```

6.3 Generate with combined inference (ML first, LLM for misses)

```
cd /Users/natarajankanakasabapathy/FECTECH/TestDataGenerator
poetry run python main.py generate --config config/Creditcard_no_rel.xlsx --output output/in
```

6.4 LLM schema enrichment (business values/rules/type hints)

```
cd /Users/natarajankanakasabapathy/FECTECH/TestDataGenerator
poetry run python main.py enrich --config config/Creditcard_no_rel.xlsx --output config/cred
```

6.5 Generate from enriched config

```
cd /Users/natarajankanakasabapathy/FECTECH/TestDataGenerator
poetry run python main.py generate --config config/creditcard_enriched.yaml --output output/
```

7) Recommended End-to-End Sequence (Copy/Paste)

```
cd /Users/natarajankanakasabapathy/FECTECH/TestDataGenerator
poetry install

poetry run python main.py lint --config config/Acct_bkng.xlsx
poetry run python main.py generate --config config/Acct_bkng.xlsx --output output/snap_v1 --c
poetry run python main.py generate --config config/Acct_bkng.xlsx --output output/snap_v2 --c

poetry run python main.py delta --config config/Acct_bkng.xlsx --previous output/snap_v1 --c
poetry run python main.py scd2 --config config/Acct_bkng.xlsx --previous output/snap_v1 --c

export SDP_LLM_PROVIDER="lm-studio"
export SDP_LLM_BASE_URL="http://localhost:1234/v1"
export SDP_LLM_MODEL="google/gemma-4-e4b"

poetry run python main.py infer-relationships --config config/Creditcard_no_rel.xlsx --confi
poetry run python main.py enrich --config config/Creditcard_no_rel.xlsx --output config/cre
poetry run python main.py generate --config config/creditcard_enriched.yaml --output output/
```

8) ER Generation (Separate Section)

8.1 ER during standard generate

```
cd /Users/natarajankanakasabapathy/FECTECH/TestDataGenerator
poetry run python main.py generate --config config/Acct_bkng.xlsx --output output/run_with_e
```

This writes ER artifacts for the run in Mermaid and DOT formats.

8.2 ER as PNG (optional)

```
cd /Users/natarajankanakasabapathy/FECTECH/TestDataGenerator
pip install matplotlib
poetry run python main.py generate --config config/Acct_bkng.xlsx --output output/run_with_e
```

8.3 ER from relationship inference workflow

```
cd /Users/natarajankanakasabapathy/FECTECH/TestDataGenerator
poetry run python main.py infer-relationships --config config/Creditcard_no_rel.xlsx --confi
```

9) Troubleshooting

infer-relationships --method llm fails with API-key/provider errors

- Confirm env vars are exported in the same terminal session:
 - SDP_LLM_PROVIDER
 - SDP_LLM_BASE_URL
 - SDP_LLM_MODEL
- Confirm LM Studio local server is running on `http://localhost:1234/v1`

Delta/SCD2 command fails due missing snapshots

- Ensure `output/snap_v1` and `output/snap_v2` exist and contain parquet files.

SCD2 produces no version changes

- Verify tracked columns are configured in workbook/YAML (`scd2_tracked_columns` or unified `cdc.track`).

Low-quality inferred relationships

- Run `record-feedback` repeatedly with SME-reviewed files; classifier improves after enough labels.
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10) Useful Test Commands

```
cd /Users/natarajankanakasabapathy/FECTECH/TestDataGenerator
poetry run pytest tests/test_rules_and_cdc.py -v
poetry run pytest tests/test_relationship_inferencer.py -v
poetry run pytest tests/test_cli_relationship_inference.py -v
poetry run pytest tests/test_config_and_parquet_flows.py -v
```